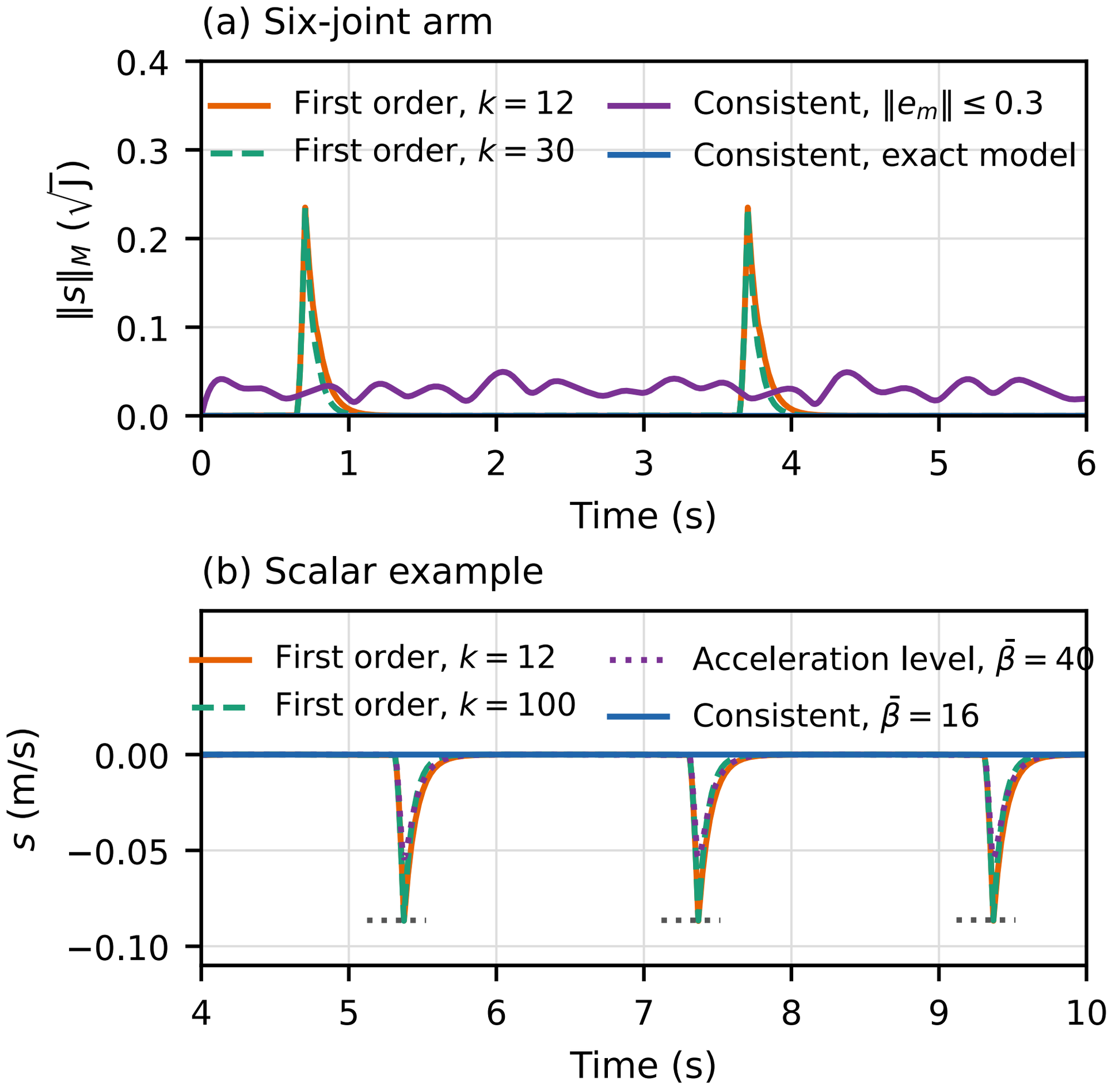
(a) Six-joint arm
First order, $k = 12$
First order, $k = 30$
Consistent, $\|e_m\| \leq 0.3$
Consistent, exact model
$\|s\|_M$ ($\sqrt{\text{J}}$)
0.4
0.3
0.2
0.1
0.0
0
1
2
3
4
5
6
Time (s)
(b) Scalar example
First order, $k = 12$
First order, $k = 100$
Acceleration level, $\bar{\beta} = 40$
Consistent, $\bar{\beta} = 16$
$s$ (m/s)
0.00
−0.05
−0.10
4
5
6
7
8
9
10
Time (s)